

Finnish Design In MoMA's Collection – A Look Through Data, Visualisation And Material Composition

Introduction

It all started with a chair. In 1942, the Museum of Modern Art in New York commissioned a birch stacking armchair (model 403) by Alvar Aalto. That was the starting point of MoMA's meticulously curated collection of Finnish design. Alvar Aalto, Finland's most renowned architect, designer, and town planner, captivated the American audience with his first-ever solo exhibition in 1938 ([McAndrew, 1938](#)). Since then, MoMA has offered a stage for over 30 artists to represent the utilitarian beauty of Finnish design at one of the United States' (and the world's) most influential cultural institutions. What we should keep in mind is that MoMA's selection is only a tiny fraction of the artistic output of those masters, especially with the majority of the designers being represented by a single artwork. This raises a question of what features of the one particular subset of Finnish design works lead it to end up in the MoMA collection? And, specifically, given the functionalist nature of modern Finnish design, what role did the physical material of the commissioned objects play? By answering these questions, we will come closer to data-backed formalisation of the MoMA curators' decision-making in regards to design items from Finland, as well as, more broadly, try to construct the "international face of modern Finnish design".

Background

The starting point of this project was a dataset published by the Museum of Modern Art in New York ([Robot, 2025](#)). It is a metadata catalogue of almost 160k artworks by over 15k artists in the possession of MoMA. The Museum acquired its first artworks

in 1929. Today, the collection contains just under 200,000 works from around the world spanning the last 150 years. The vast assembly of painting, sculpture, prints, drawing, photography, architecture, design and film represents a broad geography of over one hundred countries. Among them can be found 200+ works by forty Finnish artists. The initial thought was to utilise them all in the analysis, but later, due to the difficulty of comparing different forms of artistic expression, the preference was given to a major subset of 140 design objects.

As a baseline source of other works by designers represented at MoMA, I chose [Finna.fi](https://finna.fi) – a search service that gives access to a database of cultural and scientific materials, maintained by the National Library of Finland in cooperation with hundreds of archives, libraries and museums.

Data Collection and Preprocessing

The MoMA Collection Dataset is available for use under the CCo license and can be easily downloaded from MoMA's official GitHub repository. It comprises two sets of tabular data: 'Artworks' with metadata on every item in the museum's archive and 'Authors' with basic information about corresponding artists. Given that the provided artwork metadata is quite limited, the decision was made to enrich the dataset with images of objects, which can be retrieved from MoMA's servers via a URL included in the metadata.

When writing a Python script for downloading the images, I stumbled upon an issue with MoMA's media server blocking GET requests made by automated software. I was able to bypass the '403 Forbidden' error by adding a User-Agent header to the request, which is intended to mimic the behaviour of a browser operated by a human. Though it is a fairly regular practice, I am slightly concerned about its ethical implications. To justify this workaround, I am turning to notions that the collected images are not for commercial use and are utilised only for educational and research purposes.

The acquired data was filtered by the following parameters:

- 1) Artwork is complemented by an image, which is required for the following analysis.
- 2) Artist's nationality is Finnish.
- 3) Artwork is classified as Design.

Interestingly, some works had multiple authors. Because the intended format of the relationship was one author per artwork, the following considerations were used in dealing with such entries:

- 1) If an item is a collaboration between international artists, leave only those who are Finnish.
- 2) If an item is made by a company, it is excluded completely, because the search for related works in Finna would either inflate and skew the dataset or not give results at all. Moreover, from an artistic point of view, the question of authorship in such a case is highly debatable.
- 3) For similar reasons, if an artwork is produced in collaboration between an individual artist and a company, the authorship is attributed to the former.
- 4) Finally, if there are multiple Finnish designers listed as authors, the priority is given to the one first in the list.

Actions based on listed assumptions cannot be called perfect solutions, as they inevitably result in the loss of data. However, it led to the exclusion of only three artworks and four non-Finnish authors and companies. Such small-scale strategic changes do not harm the representativeness of individual Finnish designers and should not in any way distort the general patterns in the data.

At this stage, I have also unified the formatting of the artworks' year attribution. Specifically, the values were modified in the following way:

- 1) Estimations (e.g. 'c. 1970') were stripped of the 'c.' part.
- 2) For ranges, the lower bracket was taken.
- 3) Decades (e.g. '1950s') were rounded to a numeric part.
- 4) Non-generalisable individual cases like 'Designed 1929 (this example ca. 1936)' were handled manually.
- 5) If the year data was missing completely, the gap is filled with the median year of all artworks by the same author. However, if there is no sufficient data for

calculating the median (for instance, there is only one entry or multiple entries with missing years), then an estimate is calculated as the middle of an artist's lifetime (based on biographical data).

As with any estimates, these rough patches are further from the ideal than concrete numbers supported by factual evidence or an analysis by a specialist in the field. Nevertheless, with such an approach, we are not losing any records. And, given that the time range under scrutiny of this study spans over a century, even barbaric estimates like “1950s = 1950” still provide valuable temporal information.

Finna's database is accessible via an API ([Finna, 2025](#)). To acquire the data, I wrote a GET query which retrieves all records for physical design objects with photos, whose authors are included in a preliminarily compiled list of Finnish designers at MoMA. With over 1700 records, the baseline pool of works turned out to be significantly broader than MoMA's collection. The images were downloaded similarly to how it was done above – automatically via a provided URL. Though in this case, the program faced no server-side limitations.

The collected metadata had the same issue of missing dates, which were tackled in an analogous way. As for the authorship dilemma, for the only two entries with an author being a pair of artists, I took a different approach. I deliberately left one less represented artist as a main author to diversify the dataset and reduce the “weight” of the most popular artists.

All steps described in this section, along with corresponding code and intermediate datasets, are archived in a GitHub repository. It also includes all the collected images and constructed visualisations.

Visual Pre-Analysis

Before further analysis and modification of the data, I decided to get a general overview of the collection metadata in conjunction with its visual representation using [PIDA – Playful Interface for Digital Archives](#) – a tool for structural visualisation of visual cultural collections developed as part of my Bachelor's thesis. Initially, I planned to use it for a quick look through the images to determine the

direction of further analysis, but the process revealed some notable patterns, which were of analytical interest on their own. Moreover, it was at this stage that the initial research question (which ambitiously aimed to cover all categories of works by Finnish artists) was narrowed down to physical design, and the idea to focus on the material composition of objects was born.

Contrary to the Finna's dataset, MoMA has provided the acquisition dates for the items in its collection. And visualising them uncovers some noteworthy patterns. For example, in *Figure 1*, we can observe an iterative wave-like tendency in the acquisition of wooden and ceramic objects. It can be explained by objects being purchased/donated in batches, with it frequently being multiple works by one author, who specialises in one specific material.



Figure 1: Wooden and ceramic design objects from MoMA's collection display a repetitive pattern in their creation dates.

Overall, MoMA's acquisition pattern has been quite linear (see *Fig. 2*), with a general trend of items being created within 5–10 years before the purchase. However, there is a notable cluster in the 1980s–2000s that corresponds to a boom of enthusiasm towards works from the 1930s–1950s.

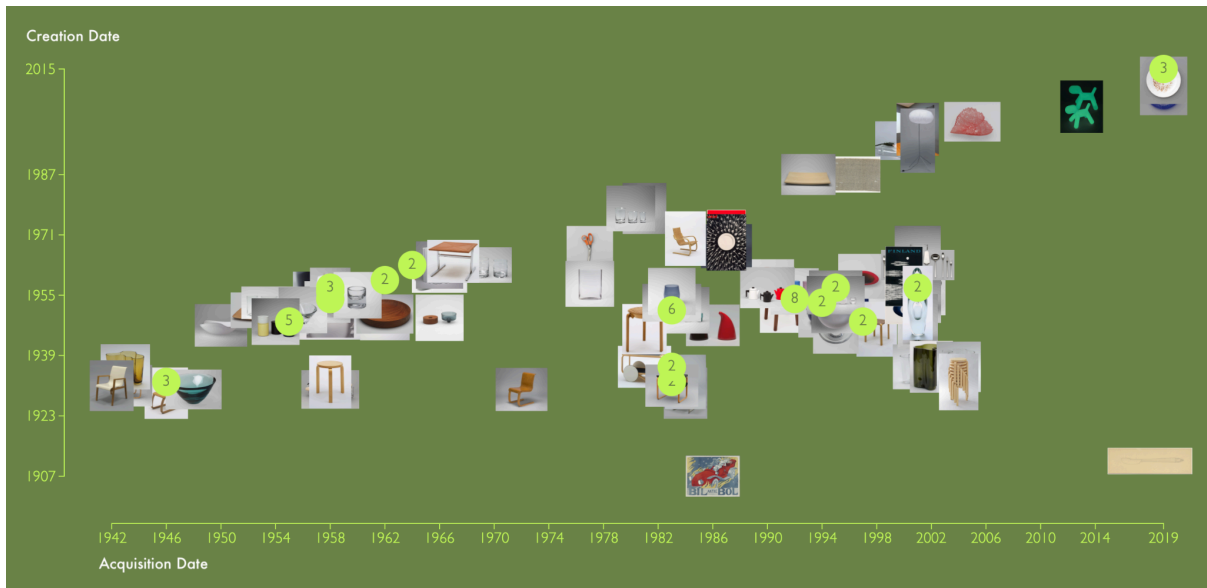


Figure 2: As a general trend, artworks were not older than 5-10 years at the moment of acquisition by MoMA, except for the period between the 1980s and the 2000s, when objects from the 1930s–1950s were in vogue.

To conclude this section, I will mention some limitations of the tool along with potential biases that can stem from them:

- 1) Limited rendering capabilities do not allow for seeing all 1700+ artworks simultaneously, with arbitrary groupings potentially hiding research-relevant patterns. Furthermore, it is generally hard to extract any visual patterns from such a large number of images.
- 2) The colour-coding feature allows for only 6 categories, meaning when applying it to display the material classification, we have to exclude the least prevalent classes. Losing them, of course, changes the composition of the data. But given the volume of information, it is highly likely that those categories would not be visible behind the major ones.

Material Classification with CLIP

One of the aims of this research is to study the collection of physical design objects – primarily, furniture, interior decor and tableware – through its material composition. MoMA’s metadata includes information on the medium of each item,

but as values in this column are too diverse and inconsistent for any kind of grouping, I had to manually condense this variety to a limited set of classes and, from it, assign a single main material to each object.

As for the Finna dataset, its metadata is fairly limited. One potential source of useful metadata is the ‘subject’ field, but with 1149 unique subjects and no explicit information about artwork medium, it cannot be used for material classification. To acquire such data, I followed the approach of applying the CLIP model ([Radford et al., 2021](#)) to an image collection for performing a zero-shot classification, as suggested by [Smits and Wevers \(2023\)](#). This method requires a predefined set of classes, which has already been compiled in the previous step. The benefits of the model are that it does not need any kind of image preprocessing and that it explicitly returns scores of proximity between images and labels, which can be turned into probabilities through a softmax function. The threshold for assigning a main material was set as 50%. Any values lower than that were treated as the model’s uncertainty, and in those cases object’s material was labelled as ‘Other’. The resulting dataset contained 275 items of this class, which accounts for roughly 17% of the records. Upon closer inspection of these objects, it is hard to fully explain the model’s output. In some cases, it is indeed hard to name only one main material of a mixed-medium item, like medals or chairs in *Figure 3*. But in other cases, the model failed to classify what is fairly easy for a human eye, like pictured glass and metal tableware.



Figure 3: Items in the collection, for which the CLIP model failed to determine the primary material they are made of.

I also observed cases of mix-ups between related classes, such as wood and paper or glass and ceramics, as illustrated in *Figure 4*. These mistakes can be attributed to both the underlying parameters of the model and the selected threshold. Perhaps, I could have set it lower to allow more items from ‘Other’ to be assigned to other classes, but it would simultaneously inflate the risk of false attributions as well as cases of two classes having equal (or almost equal) probabilities above the threshold. Moreover, the visual review suggests that the current implementation was quite

successful at correctly detecting mediums for the majority of objects, though the exact error rate could be measured only upon visual inspection by a human.

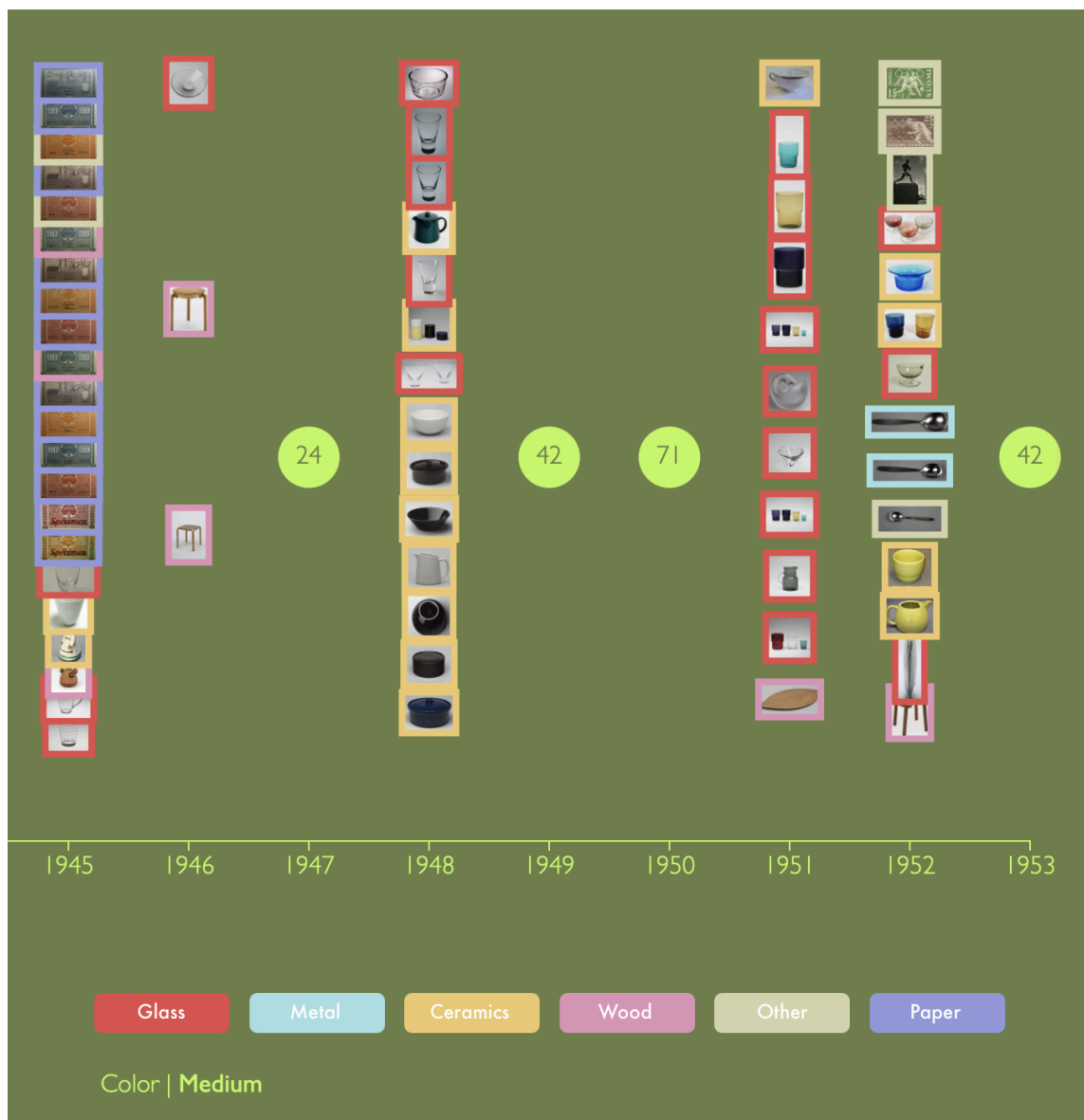


Figure 4: In most cases, the CLIP model was successful in classifying objects by their mediums. However, it is not immune to mistakes, such as attributing different classes to similar objects, for example, a series of banknotes.

Visualisation and Analysis

To mitigate the influence of the disparity in the sizes of the MoMA and Finna datasets, in all comparisons, I operated with relative, rather than absolute values.

Below are listed the key findings from visualising the relationships between two collections with different types of charts.

The first observation relates to the proportions of different materials in the two collections. As evident from the bar chart in *Figure 5*, the studied selection of artists in general employed a variety of mediums in their work (as can be judged by a more even distribution of materials in the Finna dataset). However, MoMA clearly favoured items made of glass, and, with a slightly less notable presence, wood and had no interest in textiles. [Myllyntaus \(2010\)](#) attributes the dominant role of these materials in Finnish industrial design to a) wood (and after the post-war economic development, products of the wood industry) being the biggest export commodity of Finland in that period and b) versatility of glass both as a piece of art and an everyday object, with the latter popularised, among other things, by Tapio Wirkkala's vodka bottle designs. This skew is even more visible in the detailed breakdown of each collection via treemaps in *Figures 6–7*.

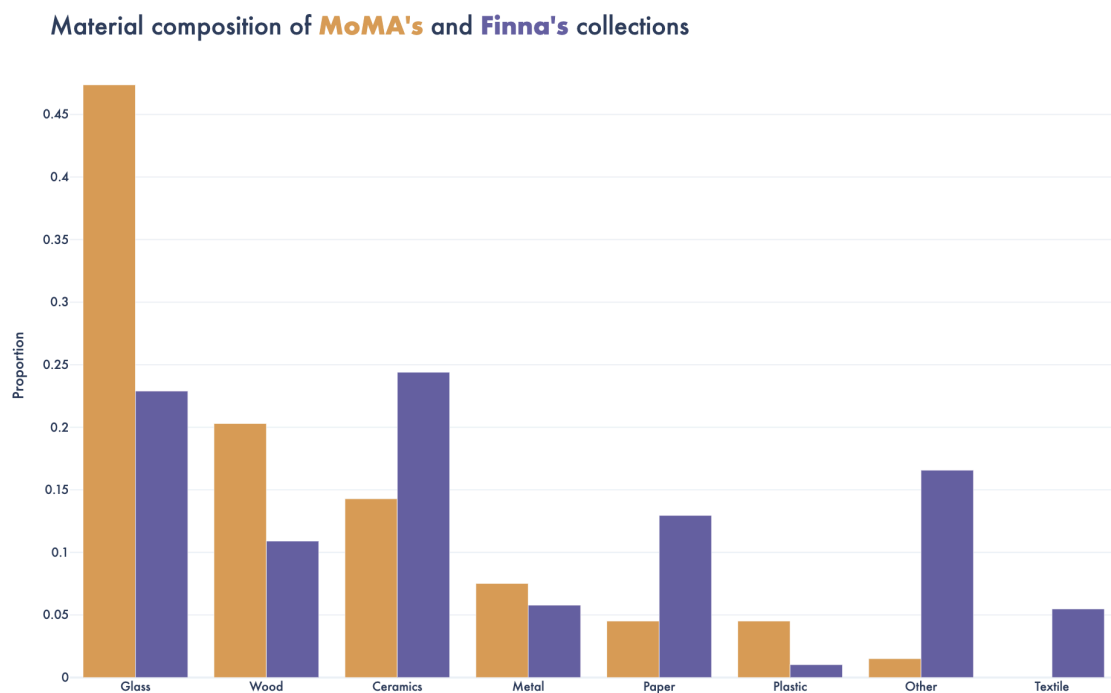


Figure 5: All mediums have different proportional representation in Finna's and MoMA's collections, with glass being the most dominant in the latter.

Materials of **MoMA**

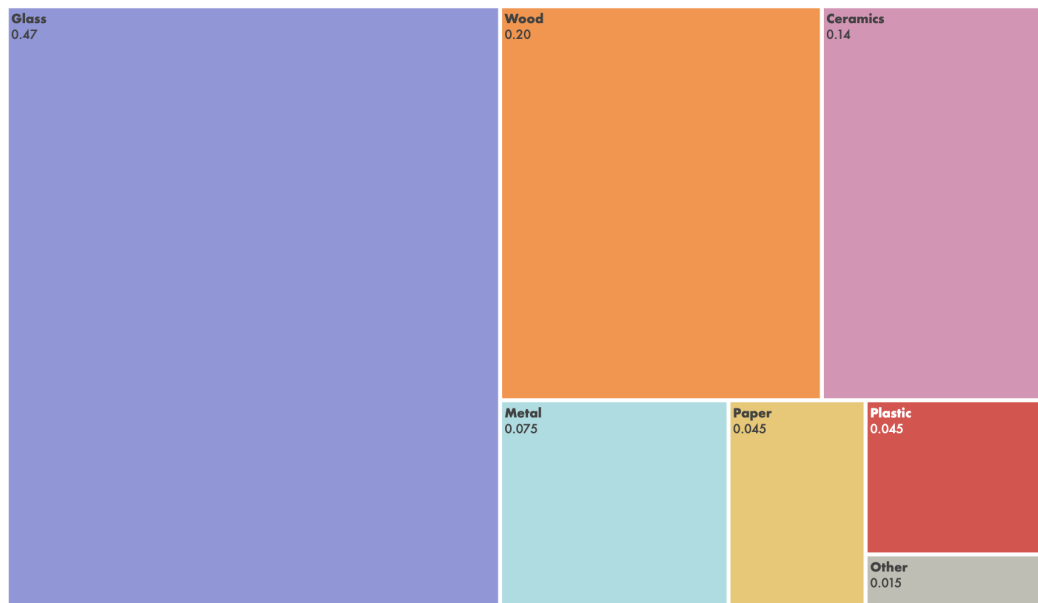


Figure 6: Distribution of artwork materials in MoMA's collection forms a strict hierarchy.

Materials at **Finna**



Figure 7: Distribution of artwork materials in Finna's collection follows a more "democratic" pattern.

Next, if we add a temporal dimension to the analysis of materials, we can see similarity in patterns between scatterplots in *Figures 8–9*. In both collections, all materials are distributed quite chaotically and more or less evenly across the studied timespan. However, there are two deviations which MoMA and Finna datasets share. Both collections welcomed plastic as a material only in the 1960s. On the contrary, wood here can be named an ‘old-fashioned’ material, with the newest wooden object in both collections created in the 1980s. Also notable are the sometimes multiple-decade-long “gaps” in the presence of some materials, the explanation of which could become a subject for research by the domain experts.

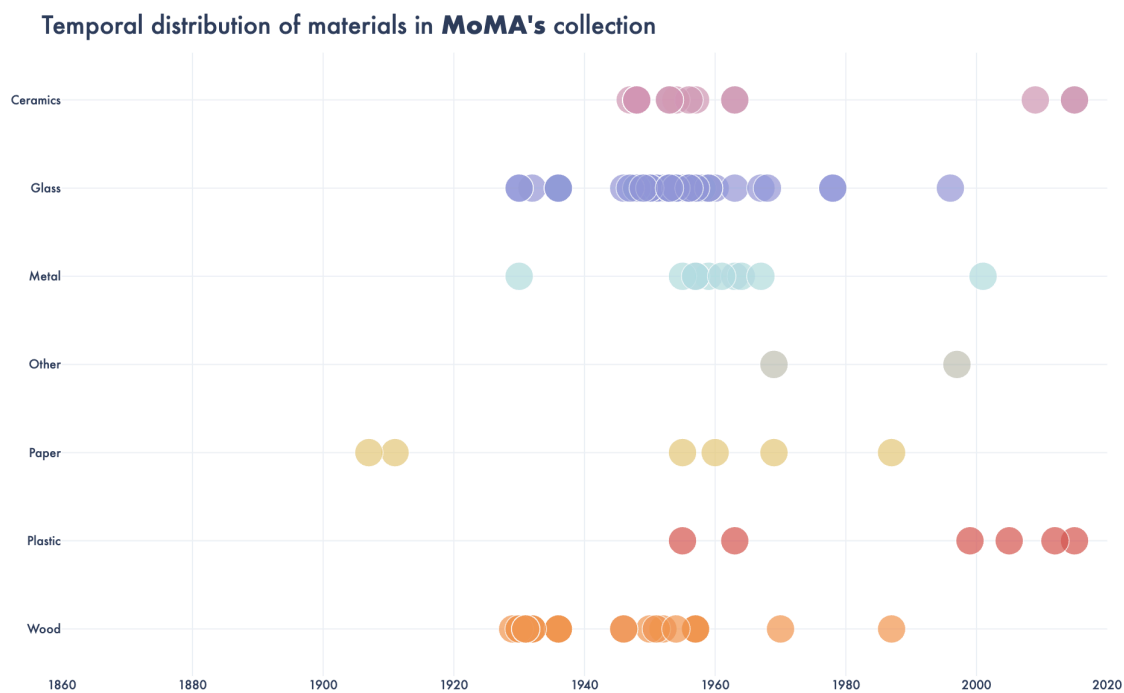


Figure 8: In MoMA’s collection, each decade has a limited set of materials present, with only the 1950s–1960s hosting all classes.

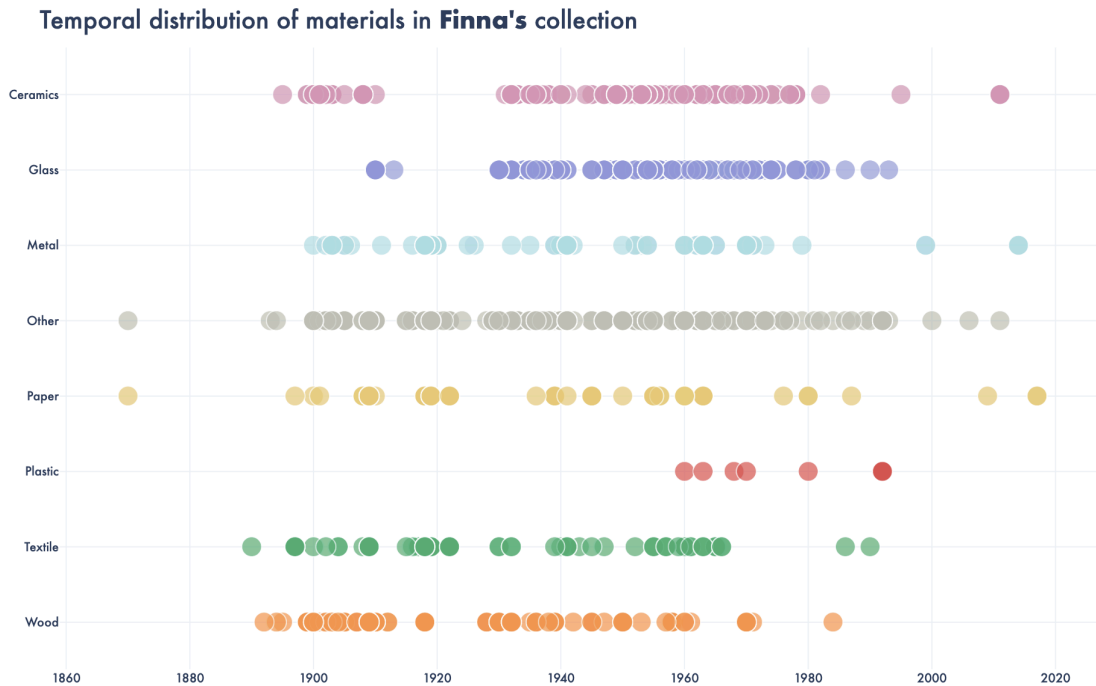


Figure 9: In Finna's collection, the materials are distributed equally along the timespan, with only plastic appearing later in the 1960s.

The major difference highlighted by the charts above is that MoMA's selection seems to be 'younger' than that of Finna. Indeed, if we look at the overall distribution of artworks across decades (see *Fig. 10*), it becomes clear that MoMA's main point of interest was works from the 1950s; almost 90% of its collection dates from the 1930s to 1960s, and there is no presence of the earlier (i.e. before the 20th century) works by the represented designers. All that despite Finna's data illustrating the Finnish design community being active throughout the whole century up until the 80s. Therefore, MoMA's choice of works cannot be attributed to the lack of objects to choose from, which leaves their artistic qualities as a primary suspect.

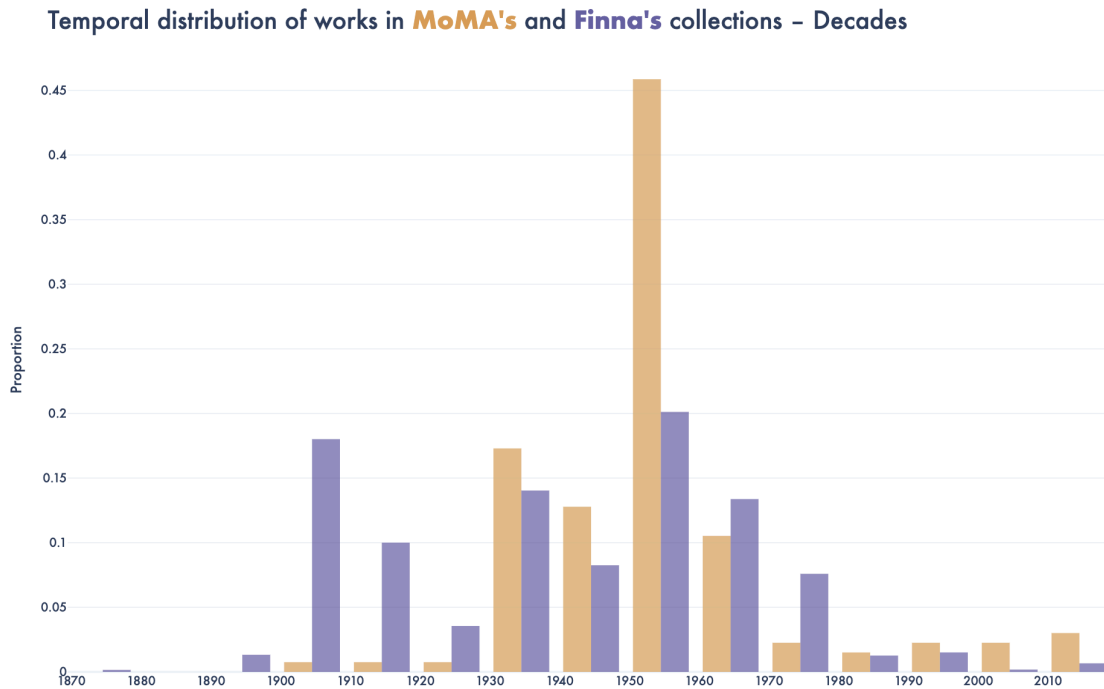


Figure 10: In Finna's collection, the distribution of artworks across decades is more even, whereas over 45% of MoMA's collection comes from the 1950s.

The comparison of representation (measured by the number of works in a collection) of individual authors brings several personas into the limelight. Architect Eliel Saarinen is by far the most prolific author in the Finna dataset, but MoMA commissioned only one of his works. Perhaps it can be explained that his career peaked at the border between the 19th and 20th centuries, way before MoMA was established, so his designs were simply not 'modern' enough for the museum's curators. On the other hand, we have international superstars Alvar Aalto and Kaj Franck, whose works dominate the MoMA's collection. Unsurprisingly, those are the only two Finnish artists for whom the Museum organised personal exhibitions ([Johnson, 1984](#); [Mount, 1992](#)). In general, all designers highlighted by the Museum actively collaborated with big Finnish manufacturers in a period of rapid growth of the consumer goods industry, which resulted in the mass production of their designs, and thus, increased visibility. Finally, the very tail of the chart in *Figure 11* displays the power of MoMA's curators in discovering talents abroad, with eight Finnish designers, whose works were commissioned by an American museum, leaving no trace in the archives of their homeland.

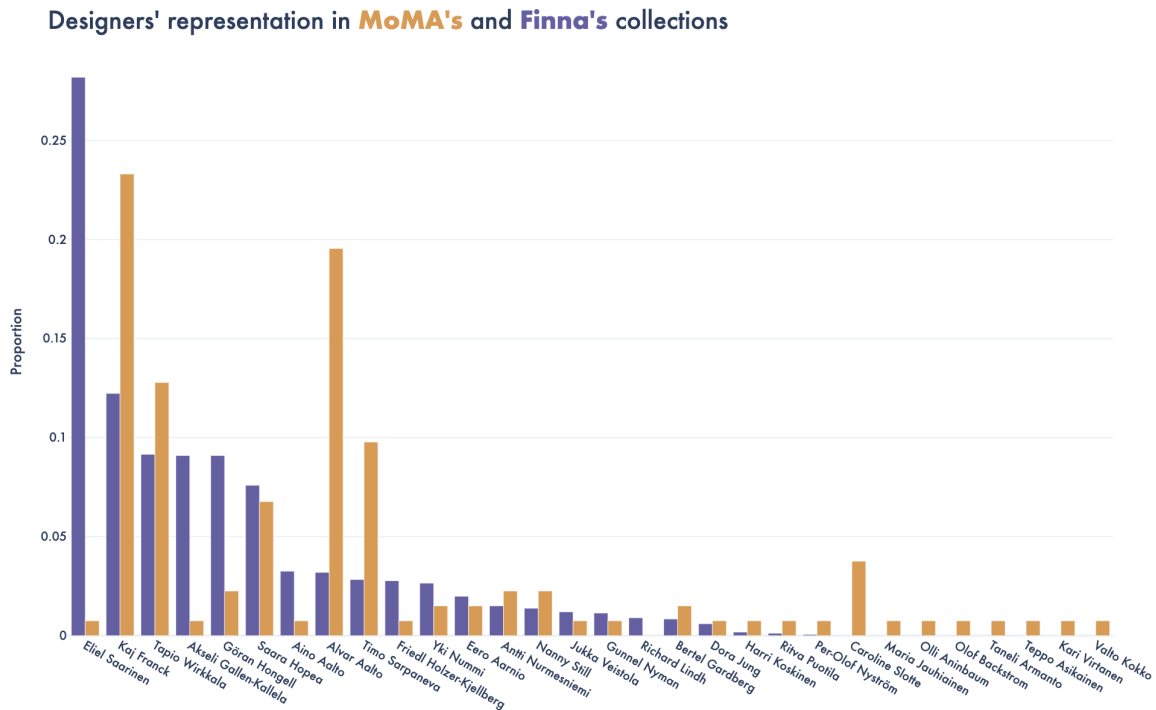


Figure 11: MoMA's selection of Finnish designers for exhibition does not indicate a connection to the scope of their presence in Finna's archives.

Conclusion

The analysis revealed that the MoMA's selection of Finnish design differs from the main pool of works in its temporal and material composition. MoMA's curators clearly favoured glass, with almost half of the items representing this medium. It could have been explained by the general tendency of Finnish designers to use this material, but Finna's data shows a way more even distribution of mediums. Therefore, we can deduce that it was a conscious curatorial decision. Evident is the general skew of MoMA for acquiring newer works, dated to the second half of the 20th century. One peculiar specificity is that almost half of MoMA's collection of Finnish design originates from the 1950s, which stands out compared to the more even temporal distribution in Finna. The attention to this 'golden decade' can be explained by it being the period of artistic activity of the most internationally renowned Finnish designers, like Alvar Aalto, Kaj Franck, Tapio Wirkkala and Timo Sarpaneva, as well as the 1952 Helsinki Olympics marking the city's debut as a cultural capital and rediscovering Finland on the world map. It was also the 1950s

when the museum was visited by the Finnish Minister of Commerce and Industry and the Finnish Ambassador to the United States, who gifted three more glass pieces to the Museum ([Museum of Modern Art, 1956](#)). So, evidently, that was a period of bonding between MoMA and Finland.

Perspective-wise, this study opens up new paths for potential analysis by specialists in art history, design theory and aesthetics. For example, using the collected data for close reading of works grouped by one material (e.g. the ‘national character’ of wood), author (e.g. the ‘feminine touch’ of Saara Hopea) or decade (e.g. the rise of the 50s and the fall of the 70s), or an investigation into the role of those mysterious designers, whose works were commissioned by MoMA, but who are not present in Finnish archives. Additionally, further narrowing down the scope of the study, i.e. working with a smaller subset of the data, would improve the readability of the visualisation in PIDA, which is more likely to yield some analytical insights into the visual content of the collections.

Concerns connected to data acquisition and processing have already been listed in the corresponding sections. Overall, the main sources of bias and areas for improvement are:

- 1) There may be an oversaturation of some individual designs, which repeatedly appear in Finna as a series of items. *Figure 12* shows an example of multiple banknote series or the same teapot shown in multiple colours. It is hard to predict the implications of this phenomenon and whether it could lead to any kind of bias in the analysis.

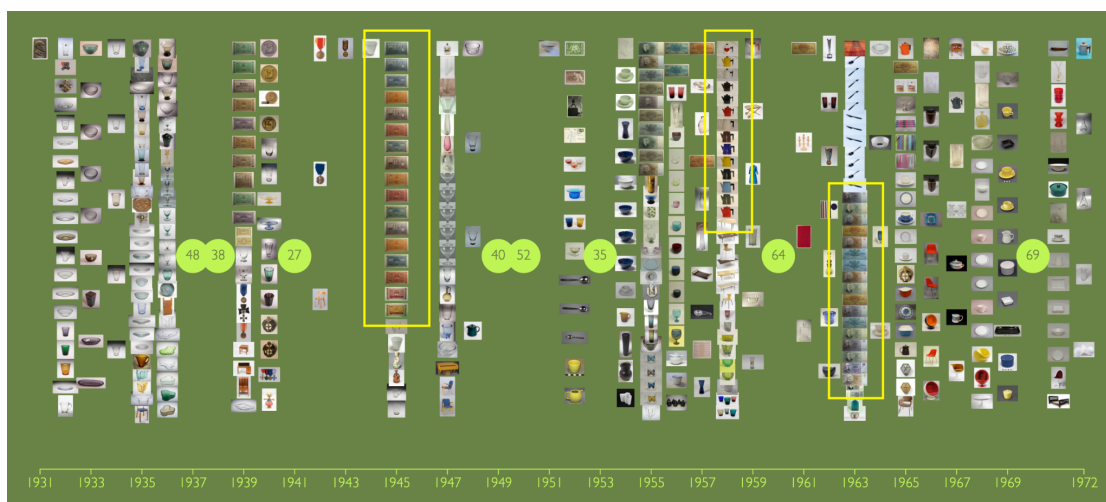


Figure 12: Clusters of variations of the same object in the Finna dataset.

- 2) Implemented workarounds for missing dates and works with multiple authors are quite blunt and simplistic. Therefore, there is still a possibility that they introduced artificial bias into the data, and, as an alternative, the complete removal of these records should also be considered.
- 3) Because data from Finna was collected automatically via an API, it lacks professional verification by a curator/museologist/art historian. So there is no certainty that the dataset does not contain wrong data/authorship attributions or wrongly linked images.
- 4) A rich source of information excluded from this study is a type of object (e.g. chairs, plates, vases, etc.), which would allow for a more fine-grained comparison. Acquiring this data would require rigorous processing (manual or with NLP tools) of artwork titles (and their 'subjects' in case of the original Finna dataset) or even the implementation of Computer Vision models for extracting this information from artwork images.

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